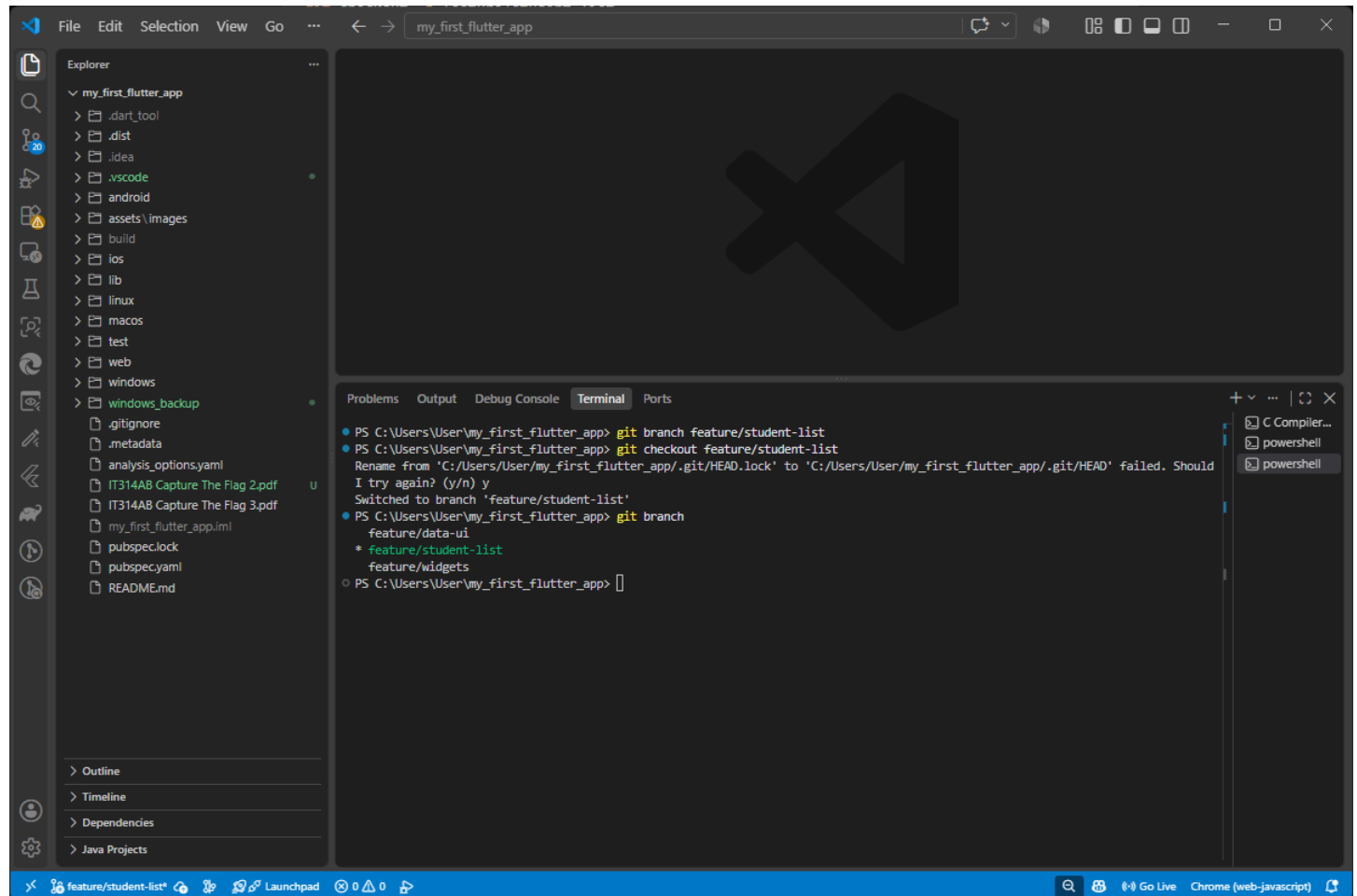


Name: Domen Jhanine B.
Subject: IT314AB

Instructor: Canque Chiara Marie C.
Date: 08/18/2026

Capture the Flag 4: The Student List

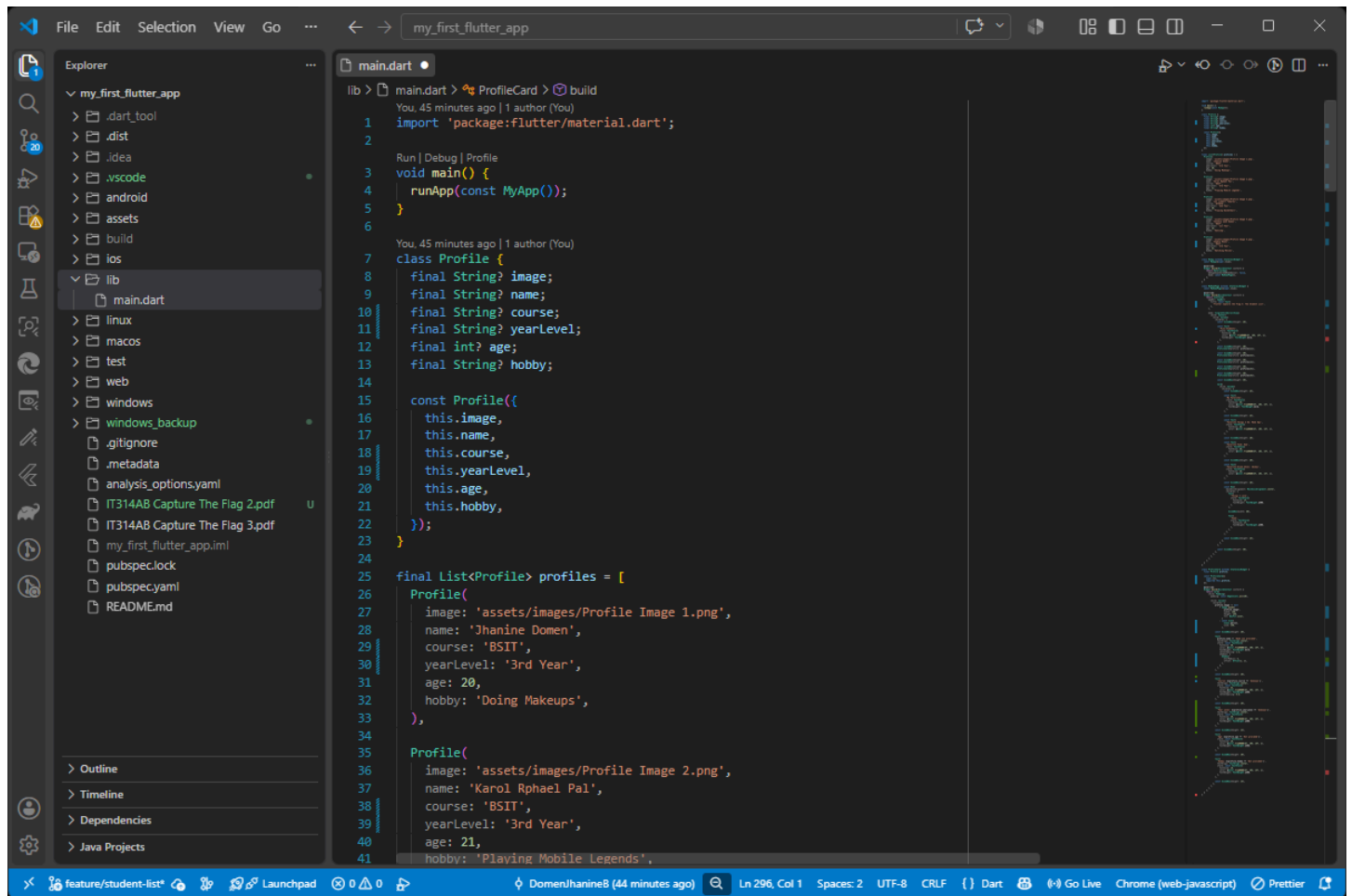
Flag 0 — Create Your Feature Branch



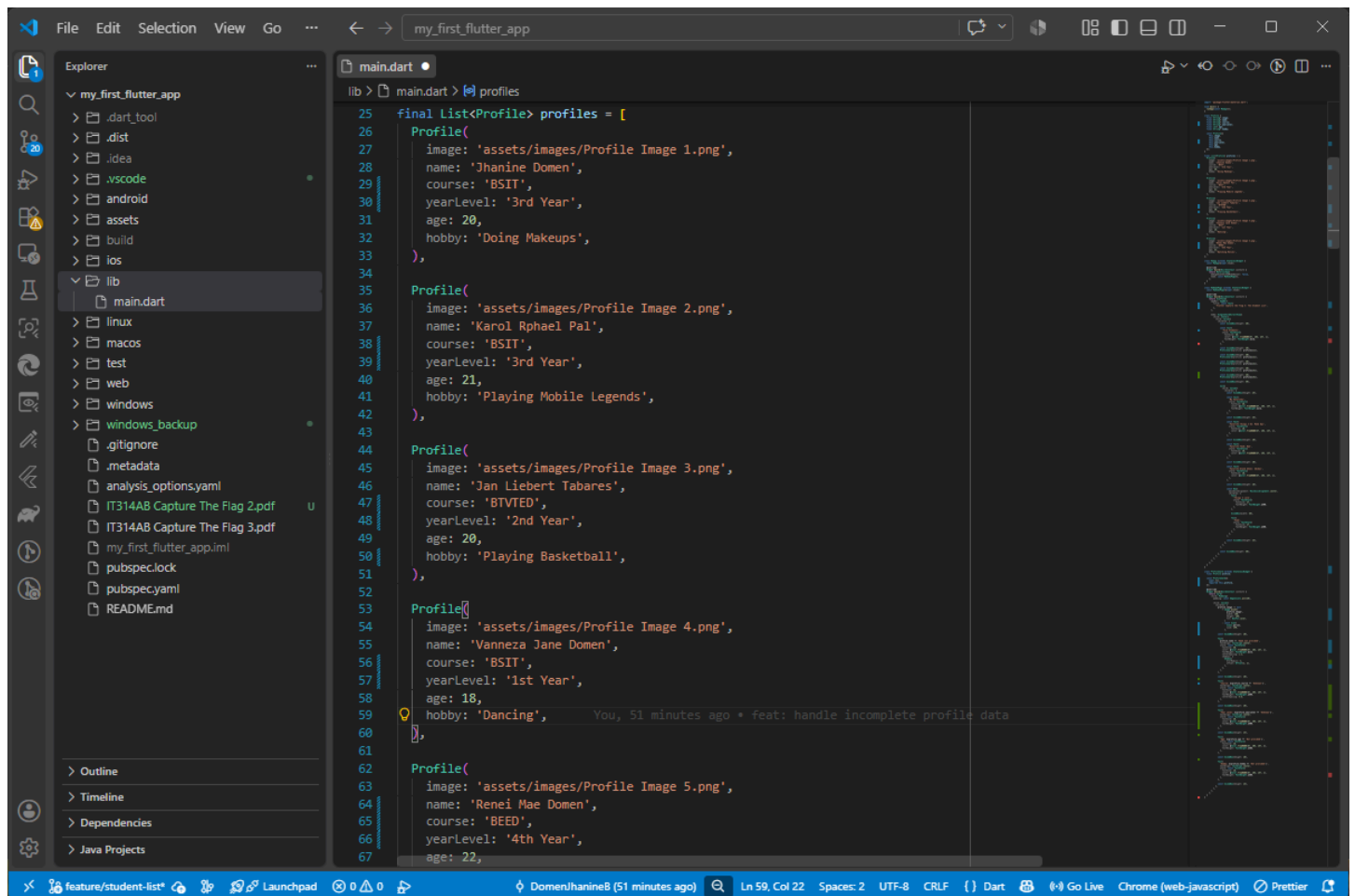
The screenshot shows the Visual Studio Code interface with a project named 'my_first_flutter_app'. The Explorer panel on the left shows the project structure, including folders like .dart_tool, .dist, .idea, .vscode, android, assets, build, ios, lib, linux, macos, test, web, windows, and windows_backup, along with files like .gitignore, .metadata, analysis_options.yaml, IT314AB Capture The Flag 2.pdf, IT314AB Capture The Flag 3.pdf, my_first_flutter_app.iml, pubspec.lock, pubspec.yaml, and README.md. The Terminal panel at the bottom shows the following commands and output:

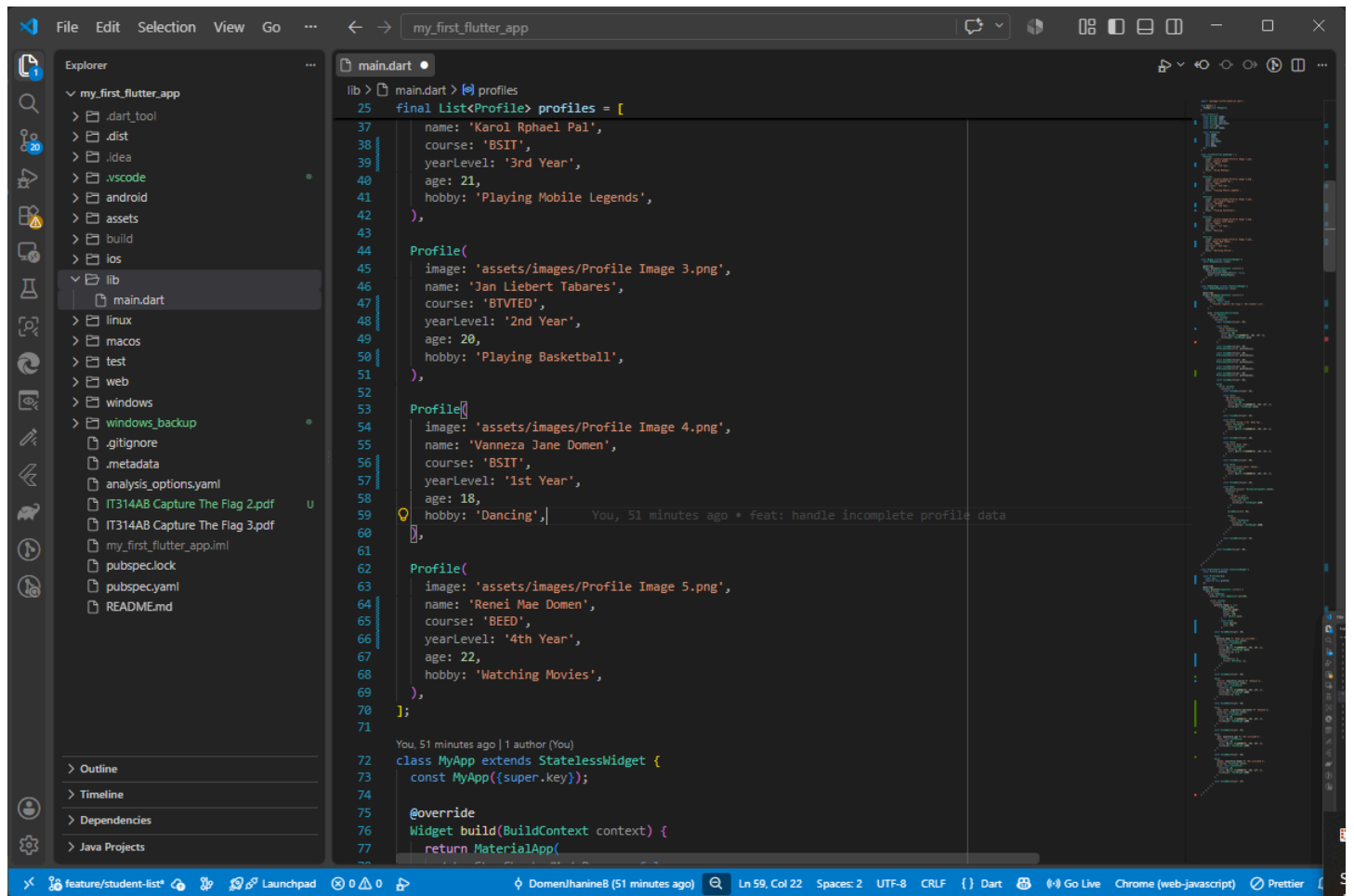
```
PS C:\Users\User\my_first_flutter_app> git branch feature/student-list
PS C:\Users\User\my_first_flutter_app> git checkout feature/student-list
Rename from 'C:/Users/User/my_first_flutter_app/.git/HEAD.lock' to 'C:/Users/User/my_first_flutter_app/.git/HEAD' failed. Should I try again? (y/n) y
Switched to branch 'feature/student-list'
PS C:\Users\User\my_first_flutter_app> git branch
  feature/data-ui
* feature/student-list
  feature/widgets
PS C:\Users\User\my_first_flutter_app>
```

Flag 1 — Build Your Student Data

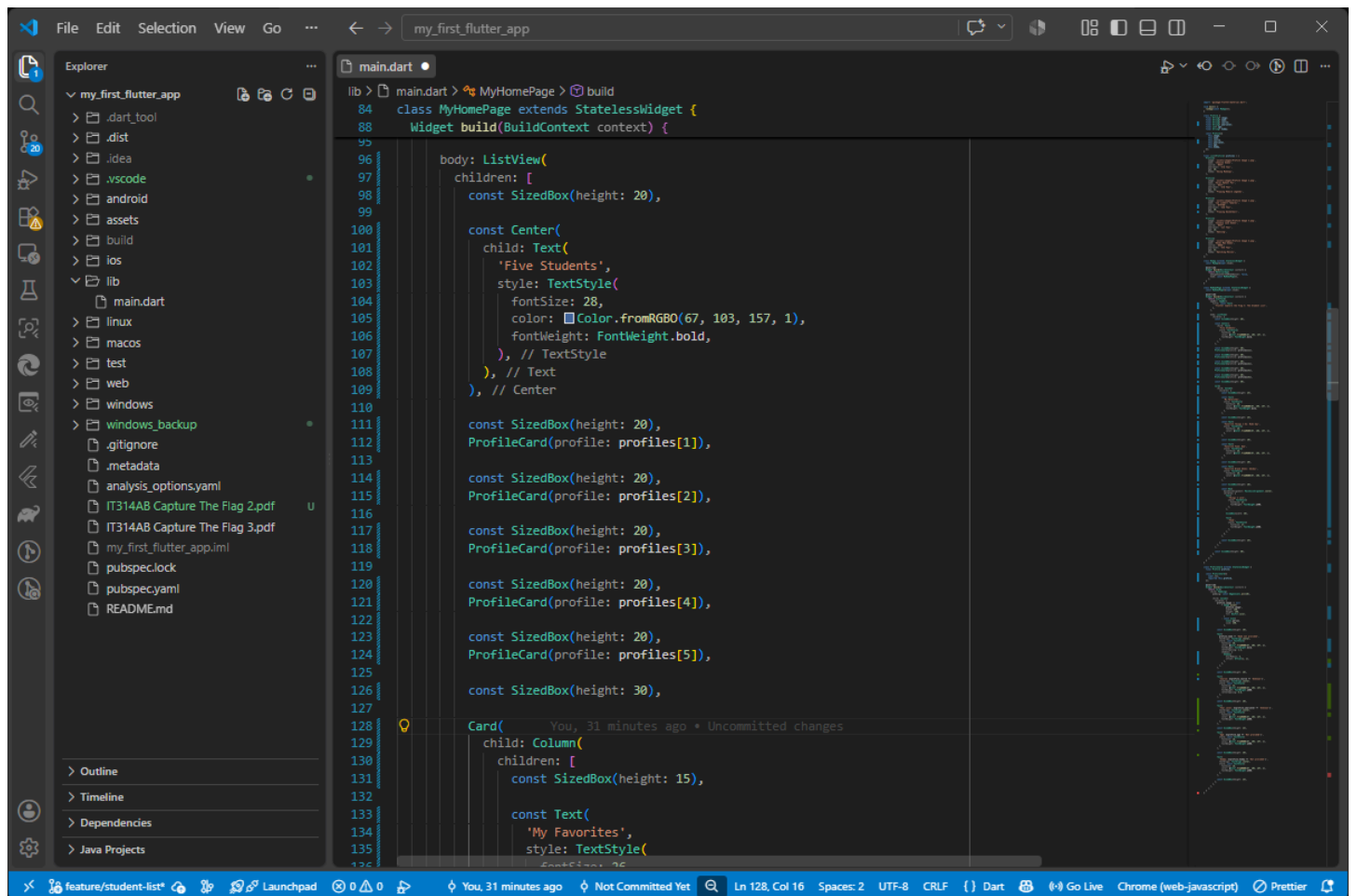


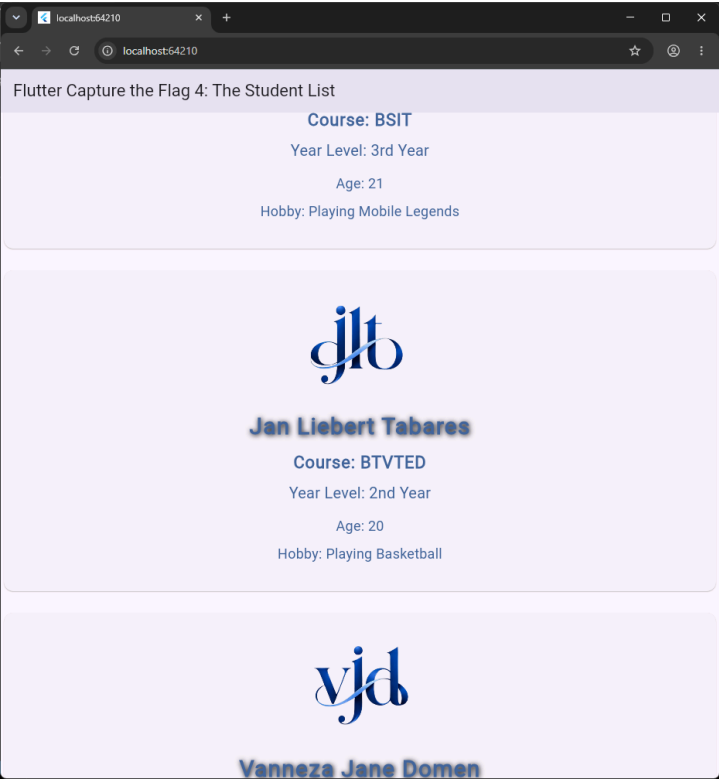
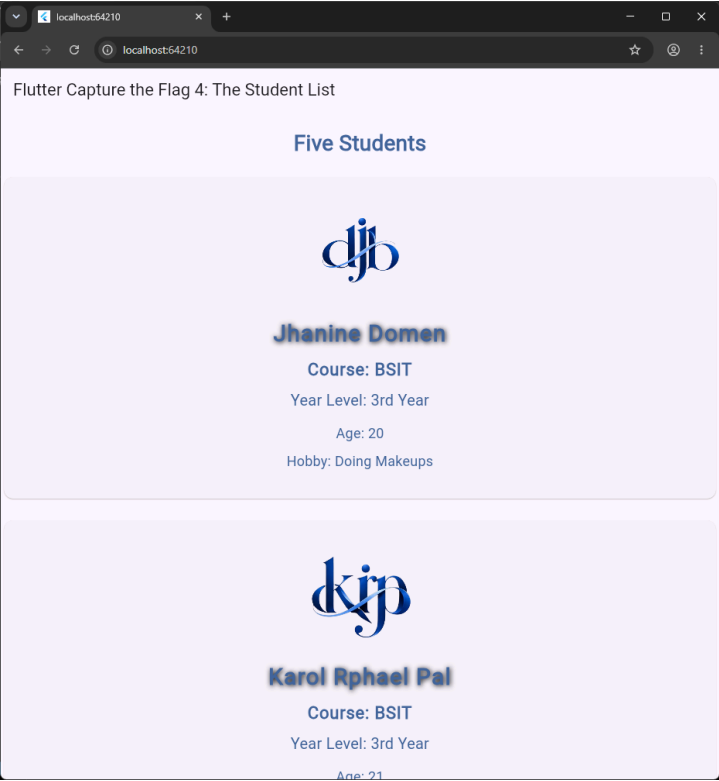
Flag 2 – The Copy-Paste Trap

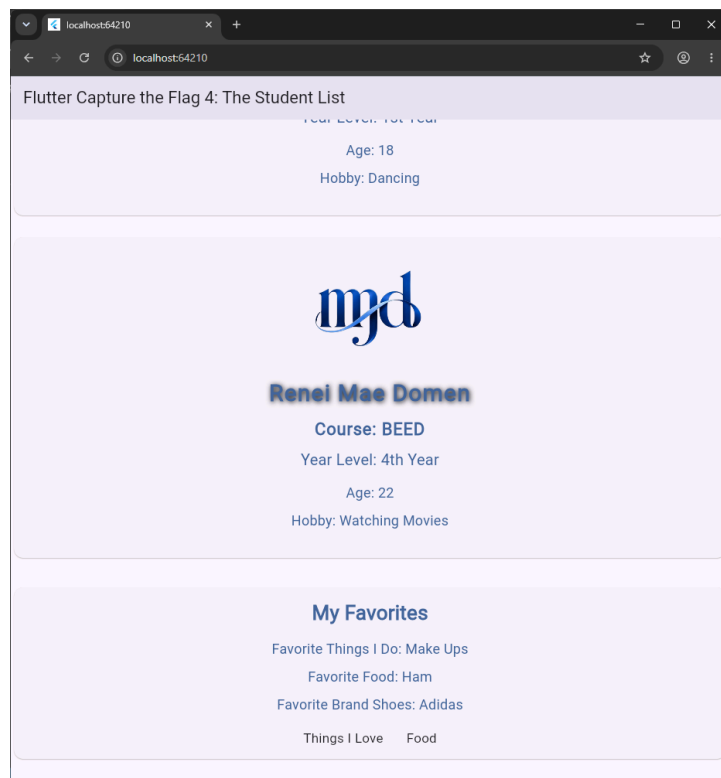
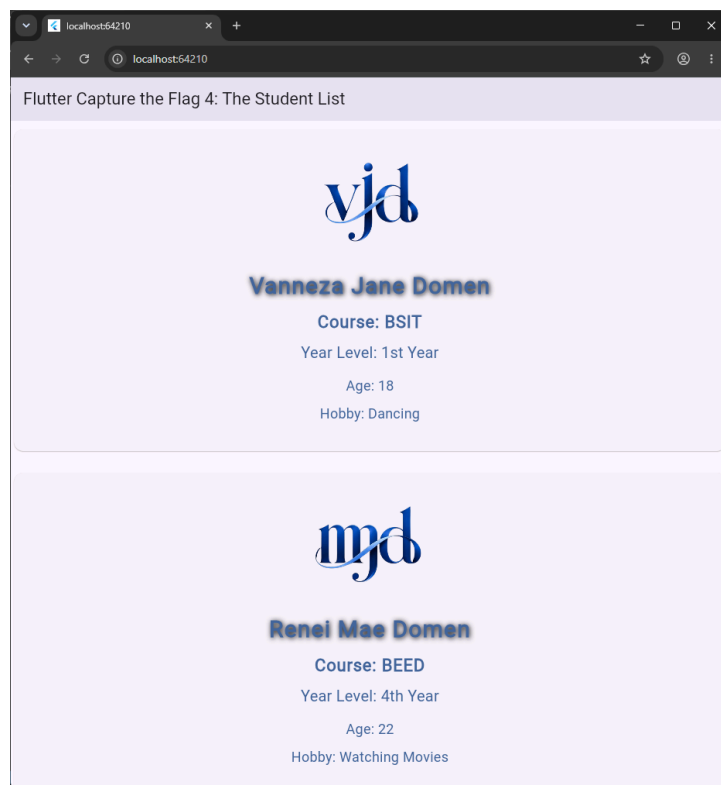




Flag 3 – Discover ListView



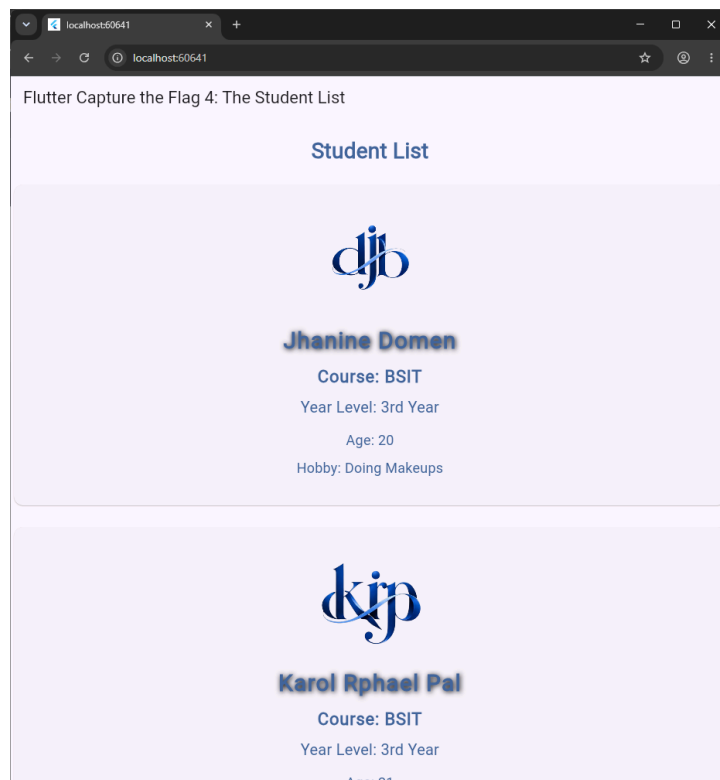


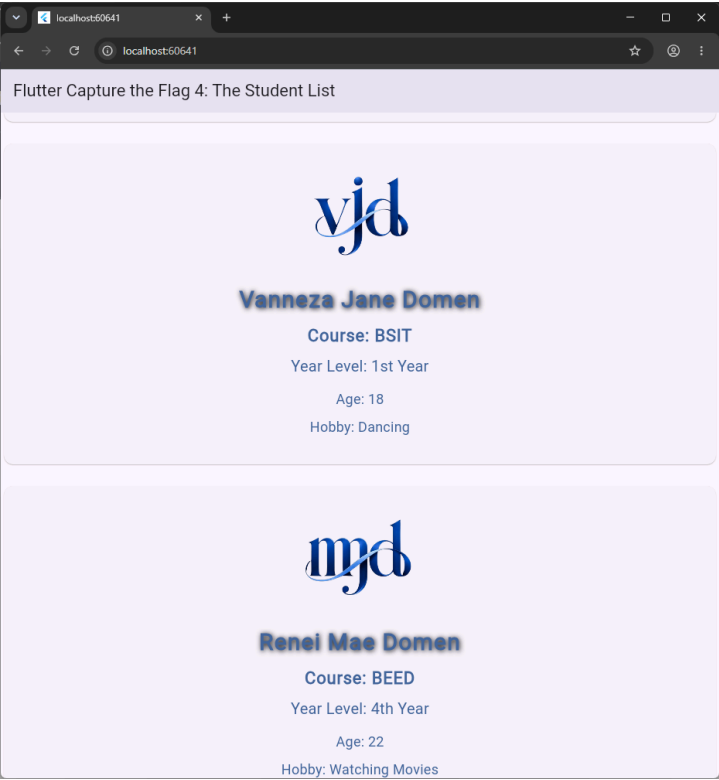
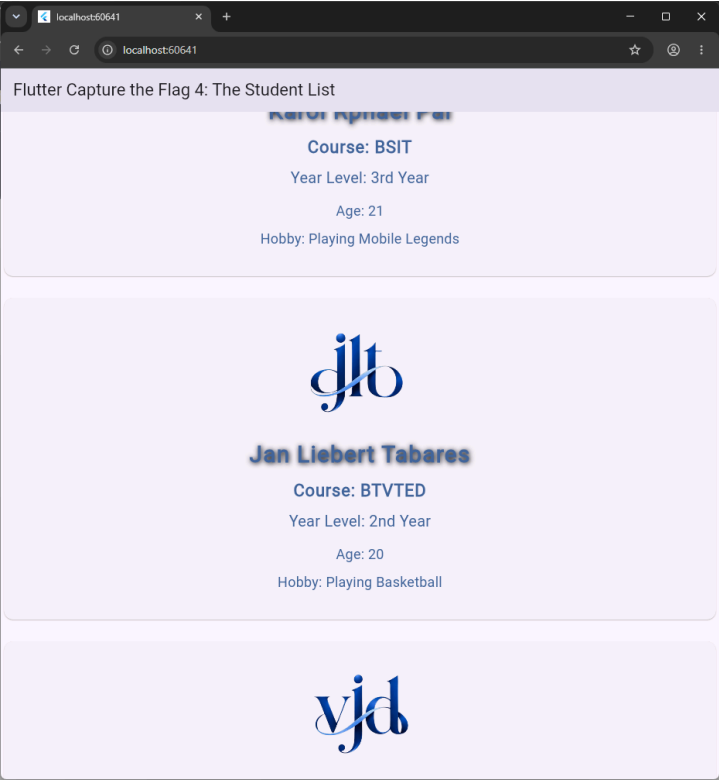


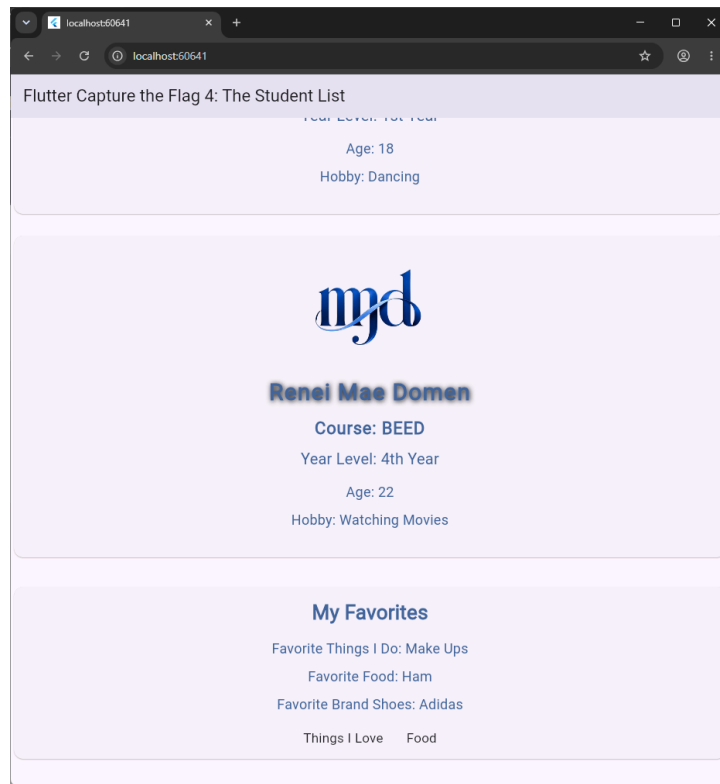
Flag 4 — Unlock **ListView.builder**

```
File Edit Selection View Go ... my_first_flutter_app

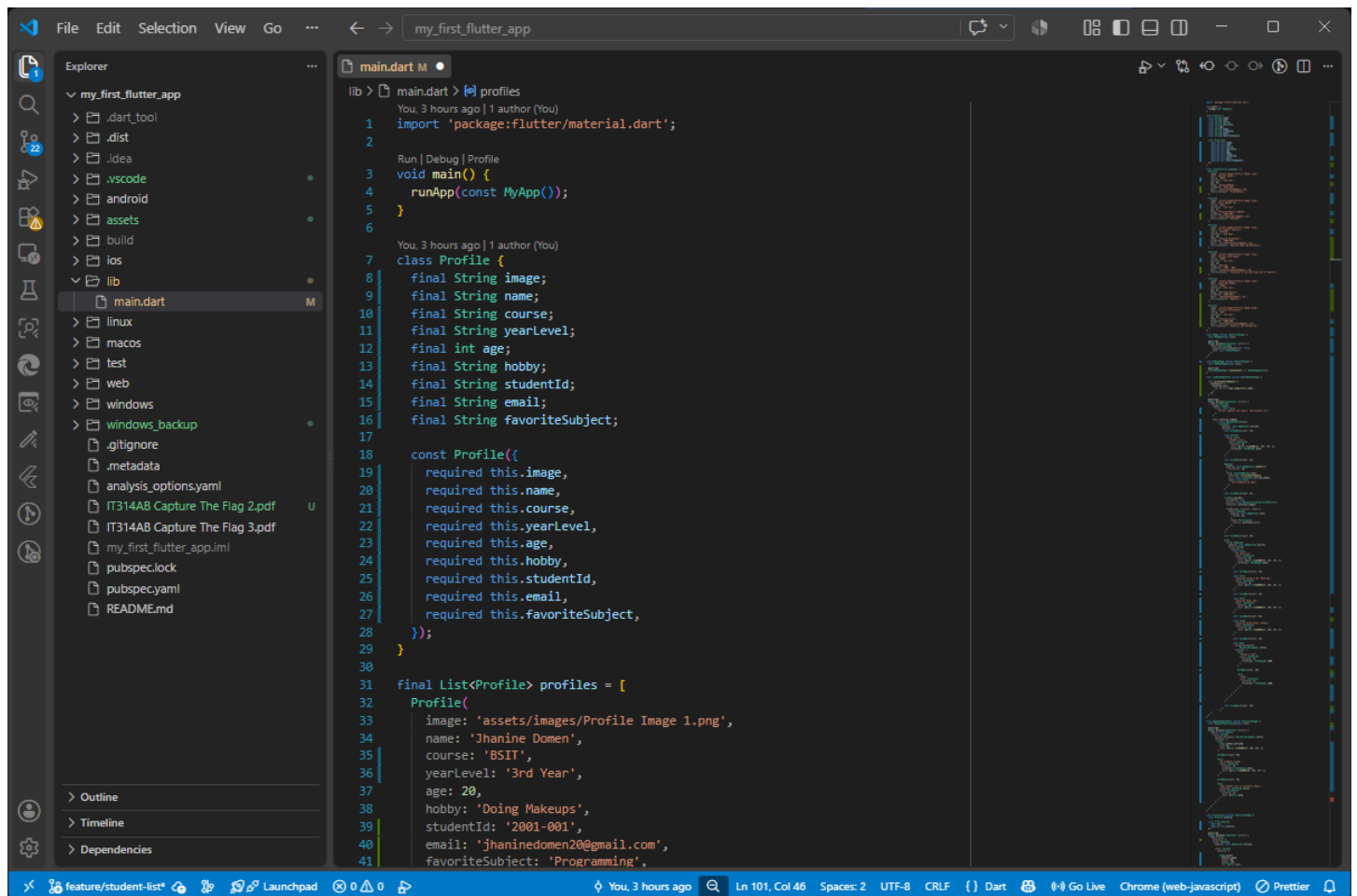
main.dart M
lib > main.dart > ProfileCard > build
84 class MyHomePage extends StatelessWidget {
88   Widget build(BuildContext context) {
101     child: Text(
102       'Student List',
103       style: TextStyle(
104         fontSize: 28,
105         color: Color.fromRGBO(67, 103, 157, 1),
106         fontWeight: FontWeight.bold,
107       ), // TextStyle
108     ), // Text
109   ), // Center
110
111   const SizedBox(height: 20),
112
113   ListView.builder(
114     shrinkWrap: true,
115     physics: const NeverScrollableScrollPhysics(),
116     itemCount: profiles.length,
117     itemBuilder: (context, index) {
118       return Padding(
119         padding: const EdgeInsets.only(
120           bottom: 20,
121         ), // EdgeInsets.only
122         child: ProfileCard(
123           profile: profiles[index],
124         ), // ProfileCard
125       ); // Padding
126     },
127   ), // ListView.builder
128
129   const SizedBox(height: 10),
130
131   Card(
132     child: Column(
133       children: [
134         const SizedBox(height: 15),
135
136         const Text(
137           'My Favorites',
138           style: TextStyle(
139             fontSize: 26,
140             color: Color.fromRGBO(67, 103, 157, 1),
141             fontWeight: FontWeight.bold,
```

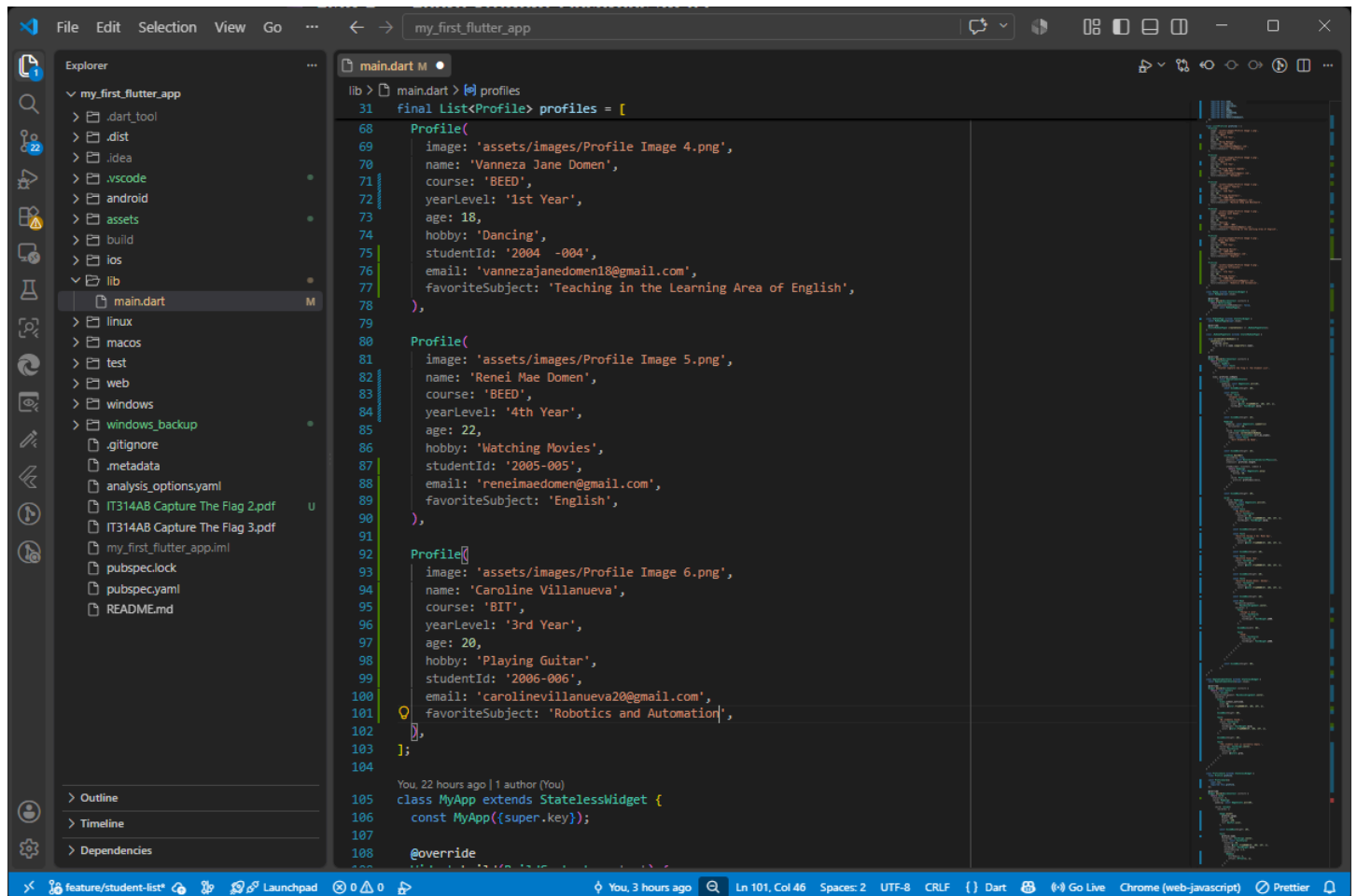
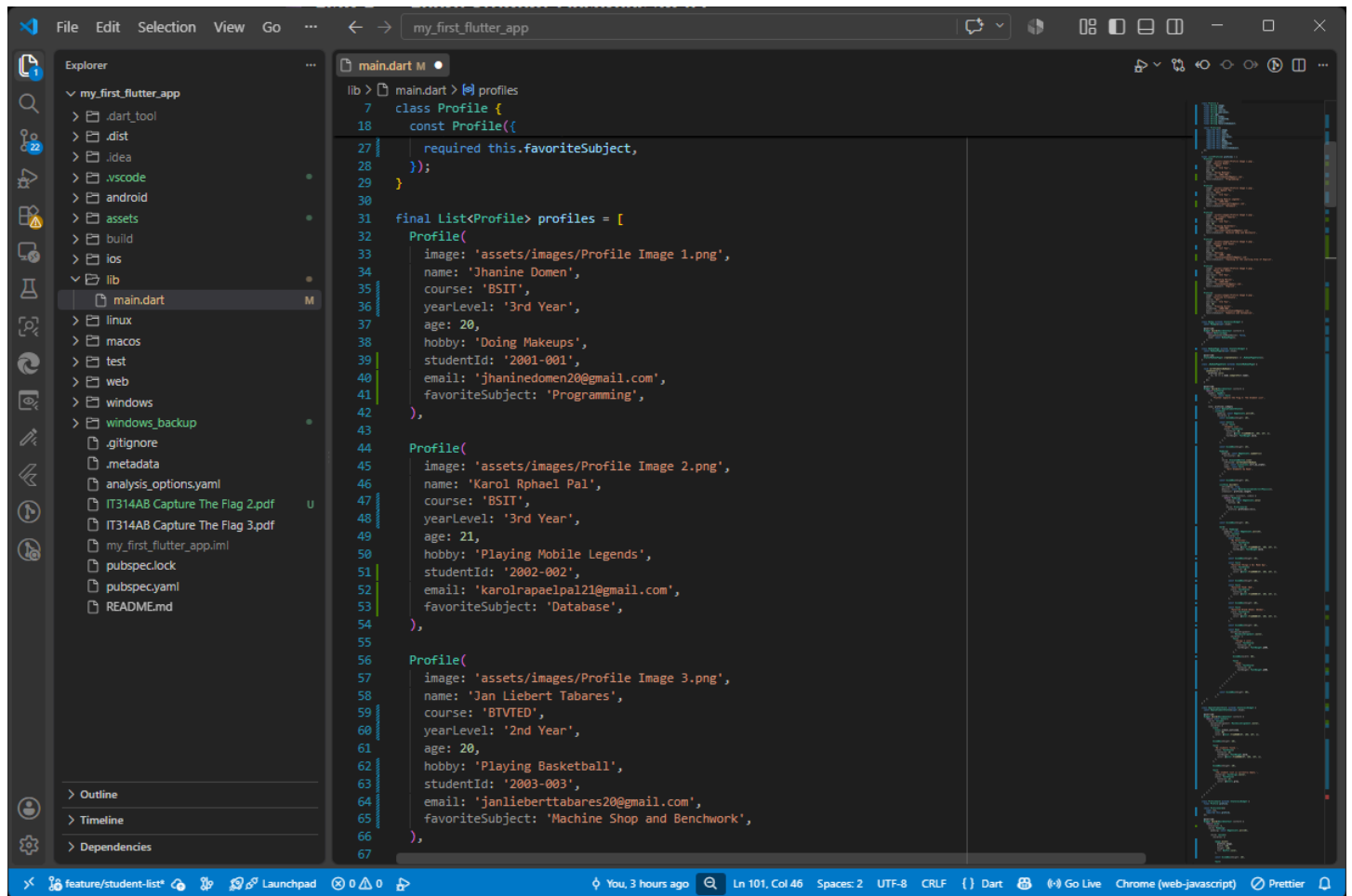






Flag 5 — Every Student Deserves an ID



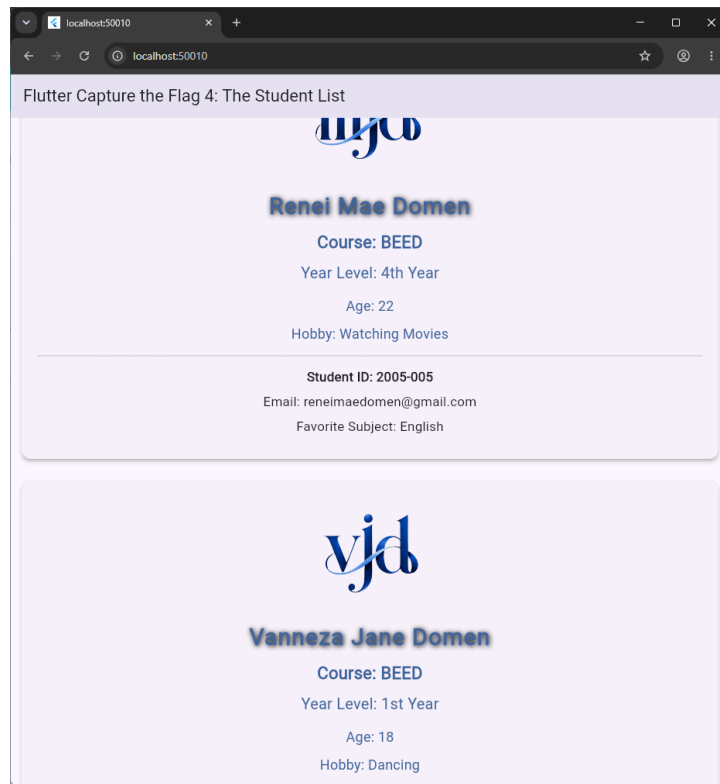


This screenshot shows the initial implementation of the `build` method in the `ProfileCard` widget. The widget is a `StatelessWidget` that builds a `Widget` from a `BuildContext`. The build method consists of several `Text` widgets and `Divider` and `SizeBox` widgets, all centered and with a specific font size and weight. The `Text` widgets display the profile's hobby, student ID, email, and favorite subject. The `Divider` and `SizeBox` widgets are used to separate the sections and provide spacing.

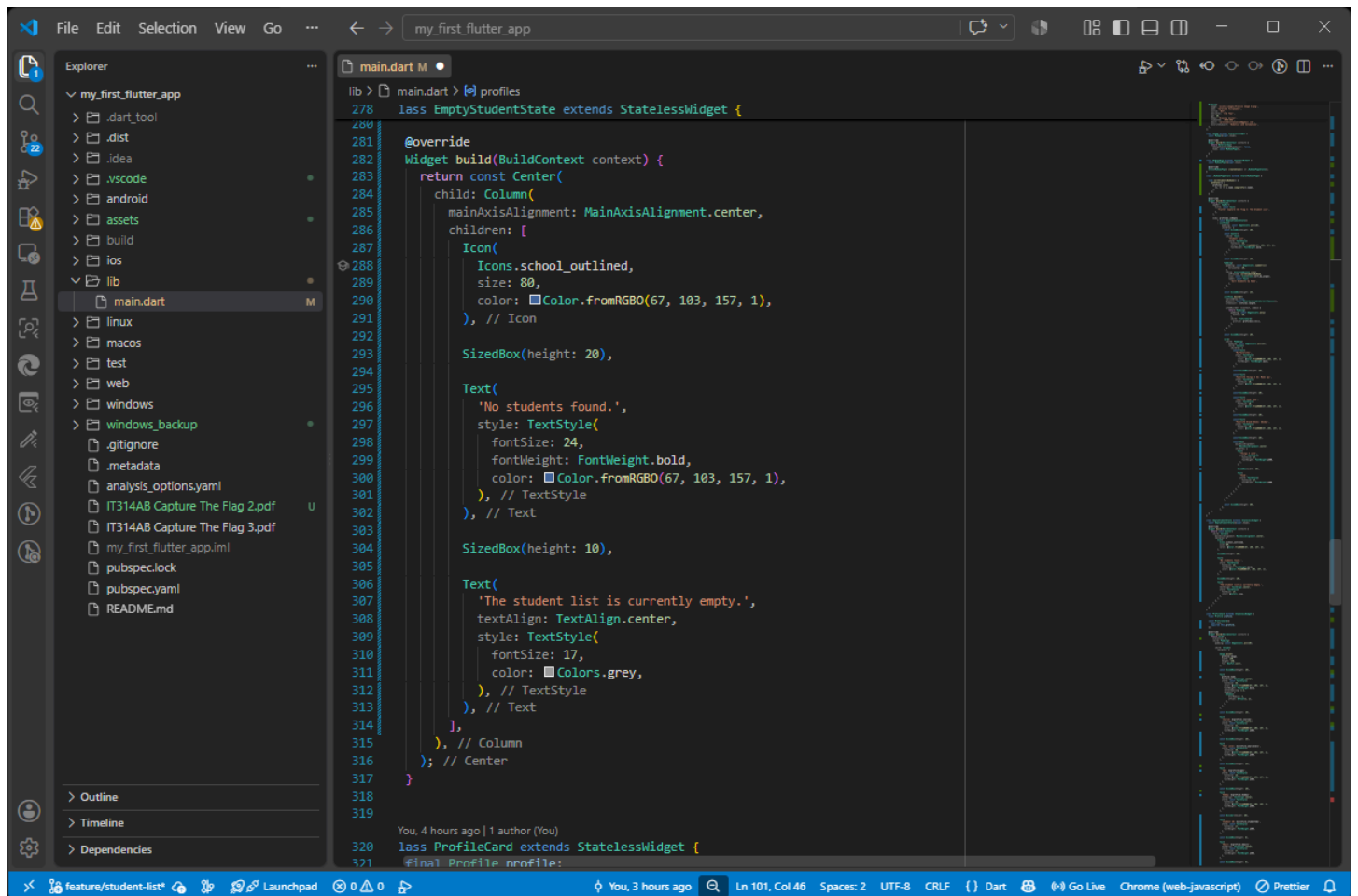
```
lib > main.dart > profiles
320 class ProfileCard extends StatelessWidget {
329   Widget build(BuildContext context) {
400
401     Text(
402       'Hobby: ${profile.hobby}',
403       textAlign: TextAlign.center,
404       style: const TextStyle(
405         fontSize: 18,
406         color: Color.fromRGB0(67, 103, 157, 1),
407         fontWeight: FontWeight.w500,
408       ), // TextStyle
409     ), // Text
410
411     const Divider(height: 30),
412
413     Text(
414       'Student ID: ${profile.studentId}',
415       style: const TextStyle(
416         fontSize: 17,
417         fontWeight: FontWeight.w600,
418       ), // TextStyle
419     ), // Text
420
421     const SizeBox(height: 8),
422
423     Text(
424       'Email: ${profile.email}',
425       textAlign: TextAlign.center,
426       style: const TextStyle(
427         fontSize: 17,
428         fontWeight: FontWeight.w500,
429       ), // TextStyle
430     ), // Text
431
432     const SizeBox(height: 8),
433
434     Text(
435       'Favorite Subject: ${profile.favoriteSubject}',
436       textAlign: TextAlign.center,
437       style: const TextStyle(
438         fontSize: 17,
439         fontWeight: FontWeight.w500,
440       ), // TextStyle
441     ), // Text
442   },
443 }
```

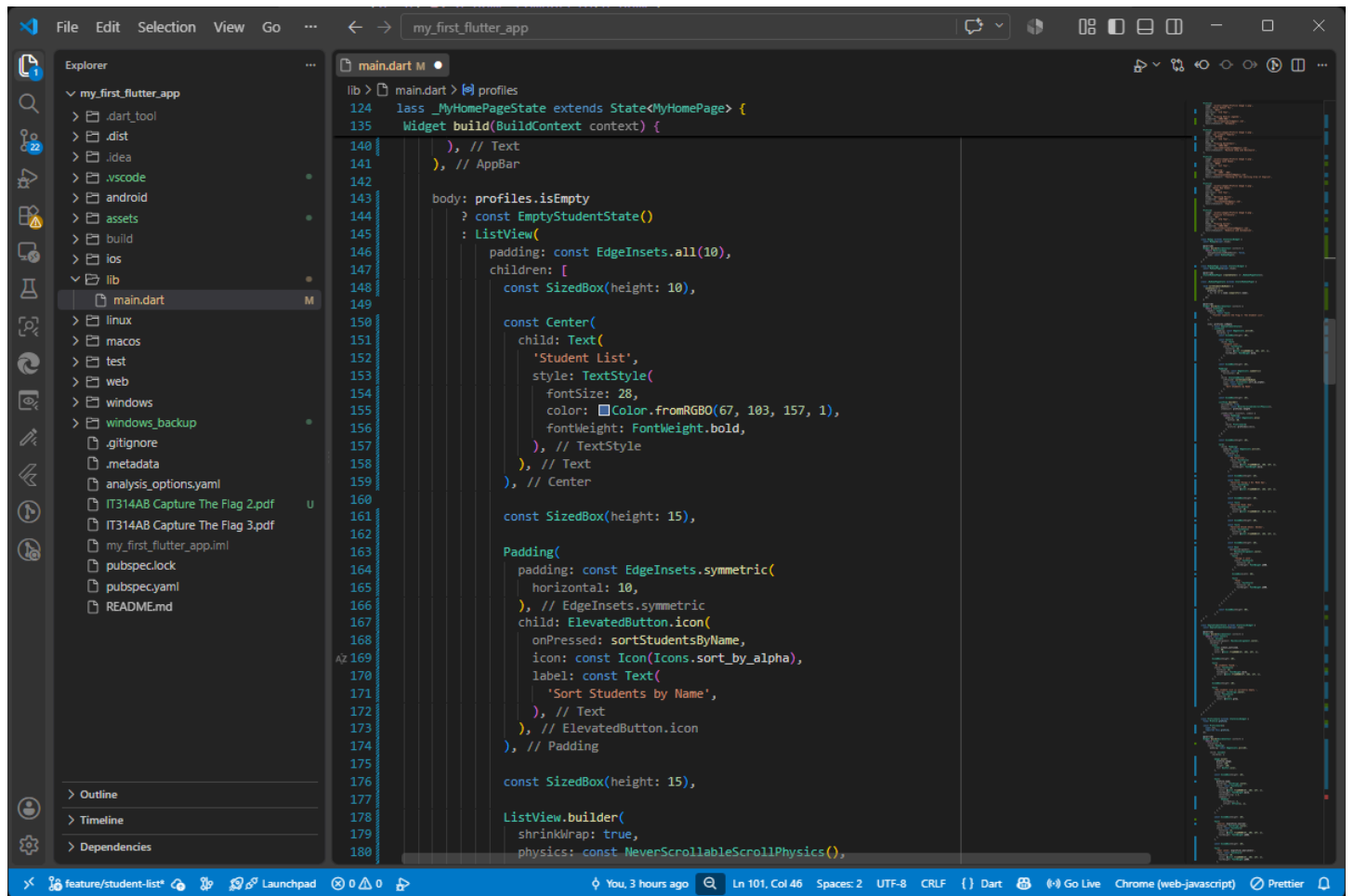
This screenshot shows the final implementation of the `build` method in the `ProfileCard` widget. The widget is a `StatelessWidget` that builds a `Widget` from a `BuildContext`. The build method consists of several `Text` widgets and `SizeBox` widgets, all centered and with a specific font size and weight. The `Text` widgets display the profile's student ID, email, and favorite subject. The `SizeBox` widgets are used to provide spacing between the sections. The final build method is wrapped in a `Column` widget with `Padding`.

```
lib > main.dart > profiles
320 class ProfileCard extends StatelessWidget {
329   Widget build(BuildContext context) {
412
413     Text(
414       'Student ID: ${profile.studentId}',
415       style: const TextStyle(
416         fontSize: 17,
417         fontWeight: FontWeight.w600,
418       ), // TextStyle
419     ), // Text
420
421     const SizeBox(height: 8),
422
423     Text(
424       'Email: ${profile.email}',
425       textAlign: TextAlign.center,
426       style: const TextStyle(
427         fontSize: 17,
428         fontWeight: FontWeight.w500,
429       ), // TextStyle
430     ), // Text
431
432     const SizeBox(height: 8),
433
434     Text(
435       'Favorite Subject: ${profile.favoriteSubject}',
436       textAlign: TextAlign.center,
437       style: const TextStyle(
438         fontSize: 17,
439         fontWeight: FontWeight.w500,
440       ), // TextStyle
441     ), // Text
442
443     const SizeBox(height: 10),
444   }, // Column
445 }, // Padding
446 ); // Card
447 }
```

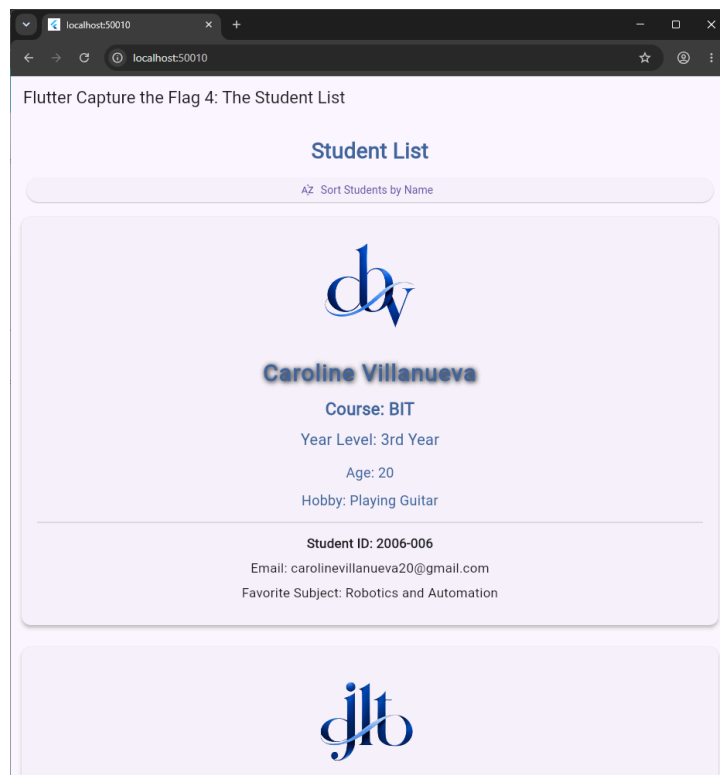


Flag 6 — Empty Classroom





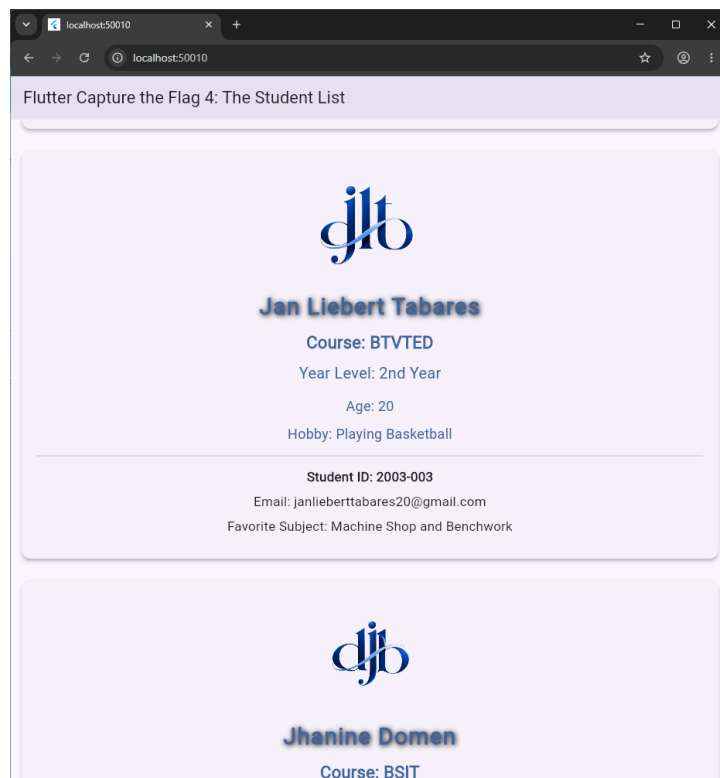
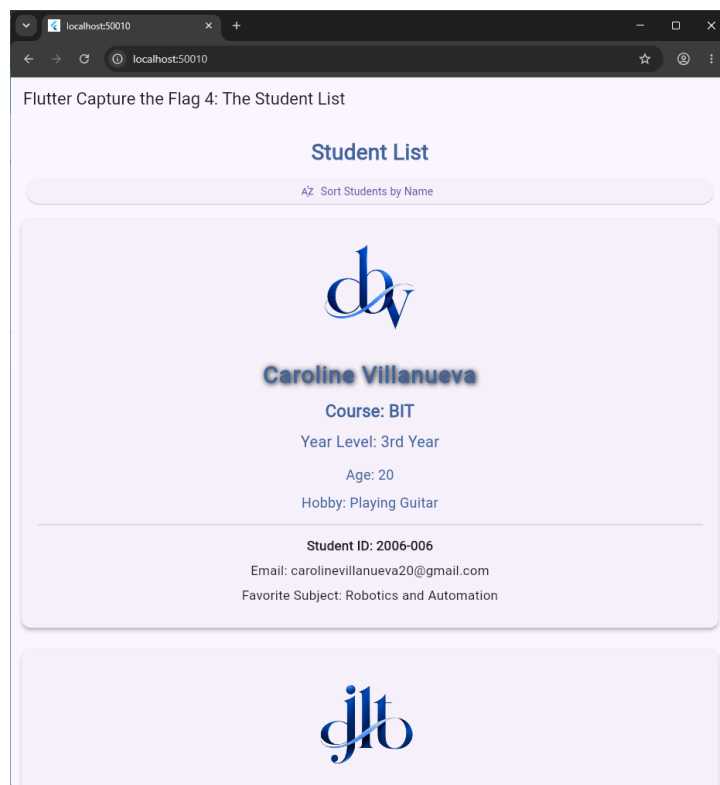
```
lib > main.dart > profiles
124 class _MyHomePageState extends State<MyHomePage> {
135   Widget build(BuildContext context) {
140   }, // Text
141   }, // AppBar
142
143   body: profiles.isEmpty
144     ? const EmptyStudentState()
145     : ListView(
146       padding: const EdgeInsets.all(10),
147       children: [
148         const SizedBox(height: 10),
149
150         const Center(
151           child: Text(
152             'Student List',
153             style: TextStyle(
154               fontSize: 28,
155               color: Color.fromRGB(67, 103, 157, 1),
156               fontWeight: FontWeight.bold,
157             ), // TextStyle
158           ), // Text
159         ), // Center
160
161         const SizedBox(height: 15),
162
163         Padding(
164           padding: const EdgeInsets.symmetric(
165             horizontal: 10,
166           ), // EdgeInsets.symmetric
167           child: ElevatedButton.icon(
168             onPressed: sortStudentsByName,
169             icon: const Icon(Icons.sort_by_alpha),
170             label: const Text(
171               'Sort Students by Name',
172             ), // Text
173           ), // ElevatedButton.icon
174         ), // Padding
175
176         const SizedBox(height: 15),
177
178         ListView.builder(
179           shrinkWrap: true,
180           physics: const NeverScrollableScrollPhysics(),
```



Flag 7 – Sort the Roll Call

```
lib > main.dart > profiles
31 final List<Profile> profiles = [
99   studentId: '2006-006',
100   email: 'carolinevillanueva20@gmail.com',
101   favoriteSubject: 'Robotics and Automation',
102 ],
103 ;
104
You, 23 hours ago | 1 author (You)
105 class MyApp extends StatelessWidget {
106   const MyApp({super.key});
107
108   @override
109   Widget build(BuildContext context) {
110     return MaterialApp(
111       debugShowCheckedModeBanner: false,
112       home: const MyHomePage(),
113     ); // MaterialApp
114   }
115
You, 3 hours ago | 1 author (You)
117 class MyHomePage extends StatefulWidget {
118   const MyHomePage({super.key});
119
120   @override
121   State<MyHomePage> createState() => _MyHomePageState();
122
123   void sortStudentsByName() {
124     setState(() {
125       profiles.sort(
126         (a, b) => a.name.compareTo(b.name),
127       );
128     });
129   }
130
131   @override
132   Widget build(BuildContext context) {
133     return Scaffold(
134       appBar: AppBar(
135         title: Text('Student List'),
136       ),
137     );
138   }
139 }
140
```

```
lib > main.dart > profiles
124 class _MyHomePageState extends State<MyHomePage> {
135   Widget build(BuildContext context) {
136     return Scaffold(
137       appBar: AppBar(
138         title: Text('Student List'),
139       ),
140       body: Column(
141         children: [
142           const SizedBox(height: 15),
143           Padding(
144             padding: const EdgeInsets.symmetric(
145               horizontal: 10,
146             ), // EdgeInsets.symmetric
147             child: ElevatedButton.icon(
148               onPressed: sortStudentsByName,
149               icon: const Icon(Icons.sort_by_alpha),
150               label: const Text(
151                 'Sort Students by Name',
152               ), // Text
153             ), // ElevatedButton.icon
154           ), // Padding
155           const SizedBox(height: 15),
156           ListView.builder(
157             shrinkWrap: true,
158             physics: const NeverScrollableScrollPhysics(),
159             itemCount: profiles.length,
160
161             itemBuilder: (context, index) {
162               return Padding(
163                 padding: const EdgeInsets.only(
164                   bottom: 20,
165                 ), // EdgeInsets.only
166                 child: ProfileCard(
167                   profile: profiles[index],
168                 ), // ProfileCard
169               ); // Padding
170             },
171           ), // ListView.builder
172         ],
173       ),
174     );
175   }
176 }
177
```



Flag 8 — Save Your Progress

```
Windows PowerShell
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (7/7), 76.61 KiB | 15.32 MiB/s, done.
Total 7 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local
objects.
remote:
remote: Create a pull request for 'feature/student-list' on
GitHub by visiting:
remote:   https://github.com/DomenJhanineB/IT314AB_Domen/
pull/new/feature/student-list
remote:
To https://github.com/DomenJhanineB/IT314AB_Domen.git
 * [new branch]      feature/student-list -> feature/student
-list
PS C:\Users\User\my_first_flutter_app> git status
On branch feature/student-list
Untracked files:
  (use "git add <file>..." to include in what will be commit
ted)
        .vscode/
        IT314AB Capture The Flag 2.pdf
        windows_backup/

nothing added to commit but untracked files present (use "gi
t add" to track)
PS C:\Users\User\my_first_flutter_app>
```